National University of Computer and Emerging Sciences

Lahore Campus

Signals and Systems (EE2008)

Date: February 25th, 2025

Course Instructor(s)

Dr. S.M. Sajid, Khalid Ijaz

Sessional-I Exam

Total Time (Hrs.): 1

Total Marks: 30

Total Questions: 3

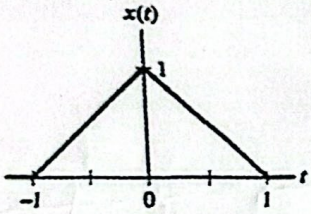
Roll No _____ Section _____ Student Signature _____

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Attempt all the questions.

CLO 1: Express different types of signals and systems.

Q1: Express the types of operations involved in the given signal x(t) for the following transformations. Also, sketch the resulting signals derived from x(t). [2.5+2.5+2.5+2.5 marks]



(a) x(3t)

(b) x(3t+2)

(c) x(−2t−1)

(d) x(2(t+2))

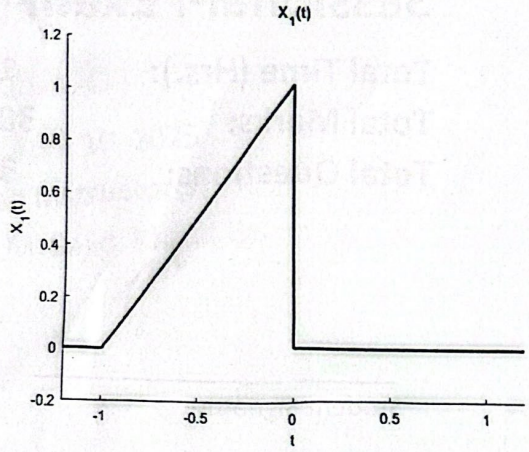
CLO 2: Apply mathematical and graphical convolution techniques to analyze continuous time and discrete time systems.

Q2: **Apply** mathematical convolutional technique to compute the convolution of $e^{\lambda t}u(t)$ with $u(t)$. [10 marks]

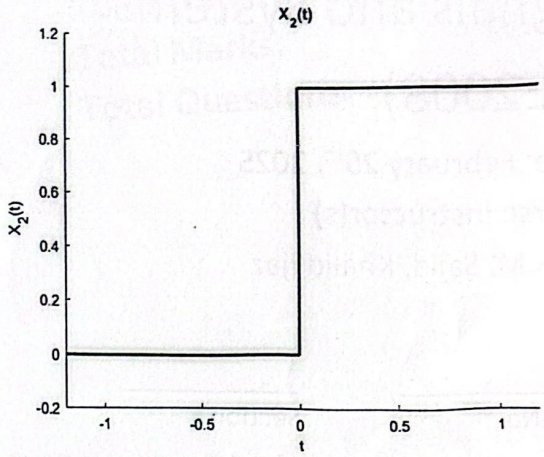
CLO 2: Apply mathematical and graphical convolution techniques to analyze continuous time and discrete time systems.

Q3: Consider the following two signals $x_1(t)$ and $x_2(t)$ as shown below. **Apply** graphical convolutional technique to compute the convolution of these two signals. [10 marks]

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(a)



(b)

$(0,1) \quad (-1,0)$

$$\frac{0-1}{-1-0} = \frac{-1}{-1} = 1$$

$y = mx + c$

$1 = 0 + c$

$c = 1$

$y = x + 1$

$x_1(t) = t + 1$

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National University of Computer and Emerging Sciences

Lahore Campus

Signals and Systems (EE2008)

Date: April 10, 2025

Course Instructor(s)

Dr. S.M. Sajid, Khalid Ijaz

Sessional-2 Exam

Total Time (Hrs.):	**1**
Total Marks:	**30**
Total Questions:	**3**

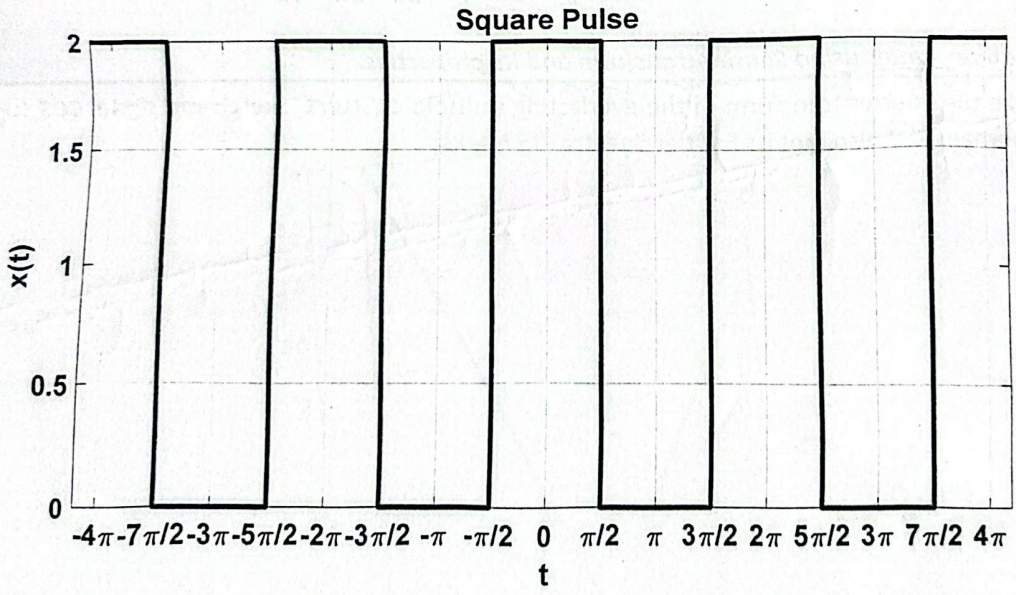
Roll No _____ Section _____ Student Signature _____

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Attempt all the questions.

CLO 3: Analyze signals using Fourier series representation.

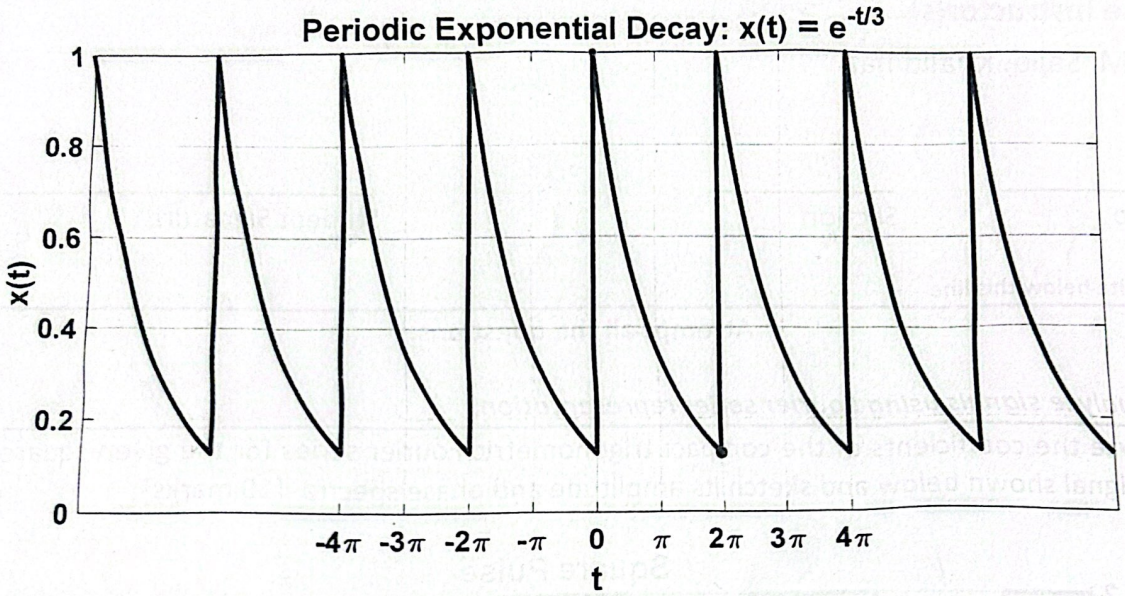
Q1: **Analyze** the coefficients of the compact trigonometric Fourier series for the given square-pulse periodic signal shown below and sketch its amplitude and phase spectra. [10 marks]



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CLO 3: Analyze signals using Fourier series representation.

Q2: **Analyze** the coefficients of the exponential Fourier series for the given exponential signal shown below. [15 marks]



CLO 4: Analyze signals using Fourier transform and its properties.

Q3: **Analyze** the Fourier transform of the everlasting sinusoid **cos $\omega_0 t$**. Sketch the signal cos $\omega_0 t$ in the time domain and also plot its Fourier Spectra. [5 marks]

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National University of Computer and Emerging Sciences

Lahore Campus

Signals & Systems (EE2008)

Final Exam

Date: May 21, 2025

Course Instructor(s)

Mr. Dr. S.M. Sajid

Mr. Khalid Ijaz

Total Time (Hrs): 3

Total Marks: 100

Total Questions: 6

[redacted] Roll No

[redacted] Section

[redacted] Student Signature

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Attempt all the questions.

CLO # 1: Express different types of signals and systems.

Q1: Express the types of operations involved in the given signal x(t) as shown in Figure 1 for the following transformations. Also, sketch the resulting signals derived from x(t). **[5+5+5+5 marks]**

a) $x(t-3)$
b) $x(t/2)$
c) $x(-t)$
d) $x(2t-4)$

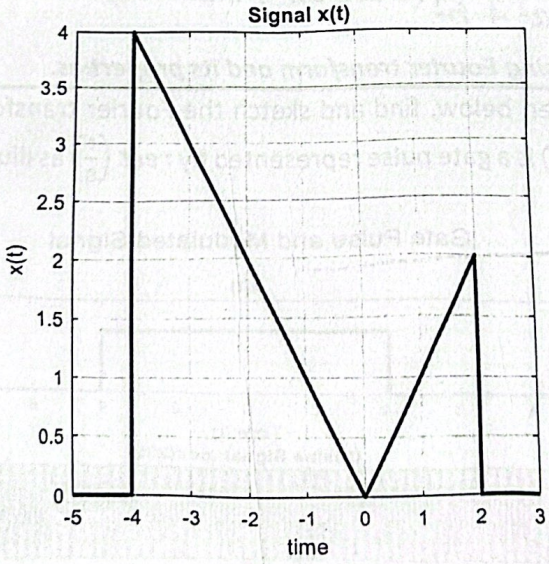


Fig-1

CLO # 2: Apply mathematical and graphical convolution techniques to analyze CT and DT systems.

Q2: Consider the following two signals x(t) and h(t) as shown in Figure 2. **Apply** graphical convolutional technique to compute the convolution of these two signals. **[20 marks]**

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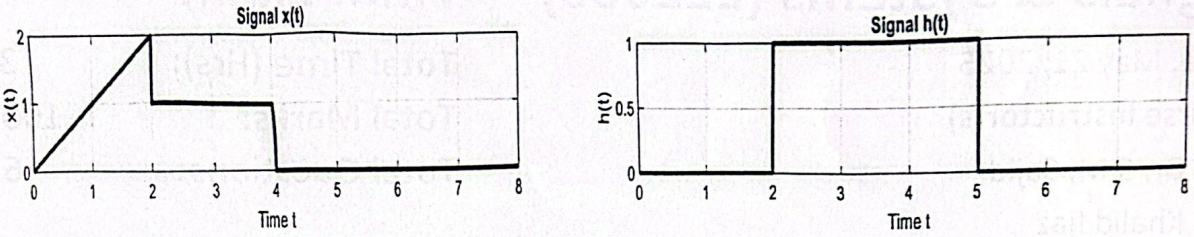


Fig-2

CLO # 3: Analyze signals using Fourier series representation.

Q3: Analyze the periodic signal shown in Fig-3, calculate all the coefficients of trigonometric Fourier series and also write the final expression of trigonometric Fourier series. **[20 marks]**

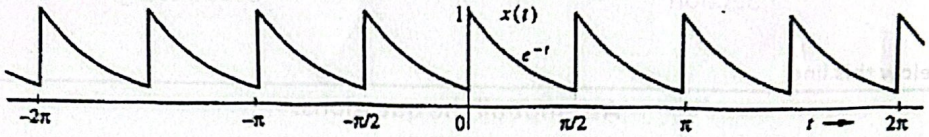


Fig-3

The following mathematical expressions can be used:

$$\int x^2 e^{ax}\,dx = \frac{e^{ax}}{a^3}(a^2x^2 - 2ax + 2)$$

$$\int e^{ax} \sin bx\,dx = \frac{e^{ax}}{a^2 + b^2}(a \sin bx - b \cos bx)$$

$$\int e^{ax} \cos bx\,dx = \frac{e^{ax}}{a^2 + b^2}(a \cos bx + b \sin bx)$$

CLO # 4: Analyze signals using Fourier transform and its properties.

Q4: Analyze the signal given below, find and sketch the Fourier transform of the modulated signal $x(t)\cos(20t)$ in which $x(t)$ is a gate pulse represented by $rect\left(\frac{t}{8}\right)$ as illustrated in figure given below. **[15 marks]**

Gate Pulse and Modulated Signal

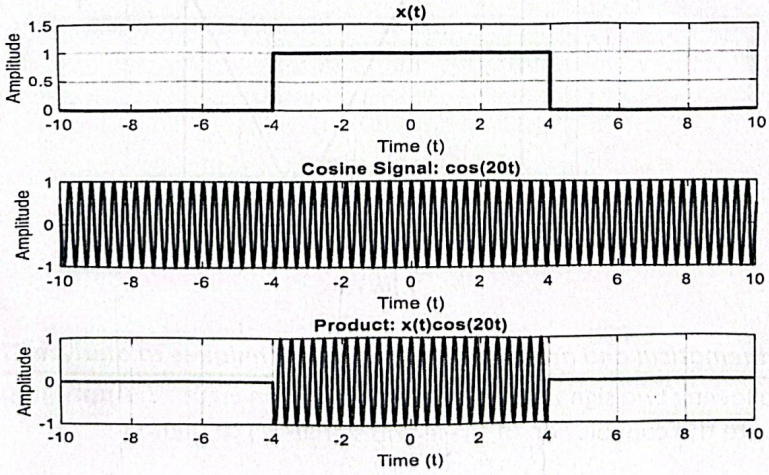


Fig-4

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CLO # 4: Analyze signals using Fourier transform and its properties.

Q5: Analyze and find the Fourier transform of $e^{-|t|}\cos(2t)$ using appropriate property of Fourier transform. **[15 marks]**

CLO # 5: Describe and use the sampling theorem.

Q6: A television signal (video and audio) has a bandwidth of 5 MHz. This signal is sampled, quantized, and binary coded to obtain a PCM signal. **[10 marks]**

a) **Describe** what the sampling rate will be if the signal is to be sampled at a rate 25% above the Nyquist rate.

b) If the Nyquist samples are quantized into 1024 levels and then binary coded, **describe** the appropriate number of binary digits required to encode a sample.

c) **Describe** the appropriate number of binary pulse rate (bit/s) required to encode the video and audio signal.

$f_s > 2 f_{max}$.

$2 \times 5 = 10\,MHz$.

$10\,MHz + 25\%$

$= 2.5 \times 10^6\,Hz$.

Sampling rate.

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